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From Data to Dynamics: A Control-Theoretic Approach for MIMO Channel Modeling

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***Abstract*—Modeling multiple-input multiple-output (MIMO) propagation in dynamic environments is inherently complex due to signal attenuation, medium variations, and user mobility. Developing accurate analytical models for such highly dynamic channel behavior is often impractical, and the computational effort grows significantly with the number of antennas. To address this, a data-driven, control-theoretic framework is proposed for modeling MIMO communication channels using only a small set of input-output measurements. By employing Observer Kalman Filter Identification (OKID) and the Eigensystem Realization Algorithm (ERA), a low-dimensional state-space model of the channel dynamics is constructed. We demonstrate that this approach accurately predicts the channel response and spectral efficiency for both One-Ring and Rician fading scenarios, showing remarkable robustness in noisy conditions.**

***Index Terms*—MIMO Communication, Data-Driven Control, System Identification, Observer Kalman Filter Identification (OKID), Eigensystem Realization Algorithm (ERA).**

I. Introduction

MODELING the multiple-input multiple-output (MIMO) channel is crucial for designing reliable wireless communication systems. However, creating accurate channel models is challenging, especially in dynamic environments where factors like signal attenuation, user mobility, and multipath scattering introduce significant complexity. While state-space methods have been explored for modeling wireless channels [1]–[3], traditional analytical models can be time-consuming to derive and often fail to capture the full range of nonlinear and time-varying behaviors. Furthermore, as the number of antennas in massive MIMO systems increases, the computational burden of these models becomes prohibitive. In this regard, data-driven system analysis has emerged as a powerful alternative for model-free characterization in wireless communications [4] and [5]. However, many existing data-driven techniques require large measured datasets, which may not be readily available in many practical scenarios. This limitation has motivated the exploration of Data-Driven Control (DDC) techniques, which are designed to build effective models from more limited data records [6]–[8].

DDC methods construct control-relevant models directly from input-output measurements, bypassing the need for a detailed physical model [9]. This approach significantly reduces modeling effort and allows for rapid adaptation to changing system dynamics. The scalability and robustness of DDC have led to its successful application in diverse fields, including feedback-controller design [10], UAV anomaly detection [11], COVID-19 forecasting [12], and power systems [13]. While DDC has been used to predict various control systems, its application to creating predictive, low-order state-space models of wireless MIMO propagation channels remains less explored.

Therefore, this paper addresses the problem of efficiently modeling MIMO channel dynamics using limited input-output data. Key contributions and novelties of the current work are summarized below.

- By leveraging Observer Kalman Filter Identification (OKID) and the Eigensystem Realization Algorithm (ERA), it provides an analytic framework to construct a low-dimensional state-space model of the wireless MIMO channel.
- An additional novelty of this work lies in how the Hankel matrix dimensions are determined for ERA in an automated, data-adaptive way. Model-order determination via singular-value thresholding of Hankel matrices is a well-established concept in system identification (e.g., see [14] and [15]); the proposed implementation tailors it to OKID-ERA for arbitrary MIMO channels in a novel, fully automated manner.

The remainder of this paper is organized as follows. Section II details the proposed modeling framework, Section III presents the simulation results and analysis, and Section IV concludes the paper.

II. Proposed Data-Driven Modeling Framework

The objective is to model the complex dynamic wireless MIMO channel as a low-dimensional linear time-invariant (LTI) state-space system. We treat the channel as a "black box", where the transmitted signals are the system's input and the received signals are its output. The proposed framework identifies this system in two steps: (1) using OKID to estimate the system's impulse response (i.e., Markov parameters) from a sequence of input-output data, and (2) using ERA to realize a minimal state-space model from this estimated impulse response.

A. State-Space Representation of the MIMO Channel

We consider a discrete-time state-space representation of the MIMO channel [3]:

$$\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k + \mathbf{B}\mathbf{u}_k, \tag{1a}$$

$$\mathbf{y}_k = \mathbf{C}\mathbf{x}_k + \mathbf{D}\mathbf{u}_k, \tag{1b}$$

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where at time step k:

- $\mathbf{x}_k \in \mathbb{C}^r$ is the unobserved internal state vector of the channel, with r being the model order.
- $\mathbf{u}_k \in \mathbb{C}^{N_T}$ is the vector of signals from the N_T transmit antennas (system input).
- $\mathbf{y}_k \in \mathbb{C}^{N_R}$ is the vector of signals at the N_R receive antennas (system output).
- $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ are the state, input, output, and feedthrough matrices, respectively, which we aim to identify.

When the system is stimulated by a discrete-time impulse input ($\mathbf{u}_k = \delta_k$) and with zero initial conditions ($\mathbf{x}_0 = 0$), the output $\mathbf{y}_k^\delta$ defines the system's Markov parameters [16]:

$$\mathbf{y}_k^\delta = \begin{cases} \mathbf{D}, & k = 0 \\ \mathbf{C}\mathbf{A}^{k-1}\mathbf{B}, & k = 1, 2, 3, \ldots \end{cases} \tag{2}$$

B. System Identification via OKID and ERA

In a practical setting, the impulse response $\mathbf{y}_k^\delta$ is not known. Instead, we have access to a set of arbitrary inputs $\mathbf{u}_k$ and their corresponding outputs $\mathbf{y}_k$.

a) Impulse Response Estimation with OKID: The Observer Kalman Filter Identification (OKID) algorithm is used to compute the Markov parameters ($\mathbf{y}_k^\delta$) from general input-output data [17]. The relationship between a non-impulse response and the impulse response is given by the convolution sum:

$$\underbrace{[\mathbf{y}_0\ \mathbf{y}_1\ \cdots\ \mathbf{y}_m]}_{\mathbf{Y}} = \underbrace{[\mathbf{y}_0^\delta\ \mathbf{y}_1^\delta\ \cdots\ \mathbf{y}_m^\delta]}_{\mathbf{Y}^\delta} \underbrace{\begin{bmatrix} \mathbf{u}_0 & \mathbf{u}_1 & \cdots & \mathbf{u}_m \\ 0 & \mathbf{u}_0 & \cdots & \mathbf{u}_{m-1} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \mathbf{u}_0 \end{bmatrix}}_{\mathbf{U}} \tag{3}$$

OKID provides an efficient method to solve for the observer Markov parameters, which are then used to recover the system's Markov parameters $\mathbf{y}_k^\delta$ [17].
Although it may seem straightforward to invert the matrix $\mathbf{U}$ to solve for the Markov parameters $\mathbf{Y}^\delta$, this approach presents several challenges:

- $\mathbf{U}$ may be non-invertible or poorly conditioned.
- For lightly damped systems, $\mathbf{U}$ becomes computationally expensive to invert due to its large size.
- Direct inversion does not optimally filter noise.

To address these issues, OKID can be leveraged to introduce an optimal observer system [18]:

$$\hat{\mathbf{x}}_{\mathbf{k+1}} = \underbrace{(\boldsymbol{A} - \boldsymbol{K_f C})}_{\bar{\boldsymbol{A}}_{\boldsymbol{d}}} \hat{\mathbf{x}}_{\mathbf{k}} + \underbrace{[\boldsymbol{B} - \boldsymbol{K_f D}, \boldsymbol{K_f}]}_{\bar{\boldsymbol{B}}_{\boldsymbol{d}}} \begin{bmatrix} \mathbf{u_k} \\ \mathbf{y_k} \end{bmatrix} \tag{4}$$

where, A, B, C, D are the discrete-time system matrices and K_f is the Kalman filter gain matrix.
The OKID algorithm proceeds as follows [18]:

1) Choose the number of observer Markov parameters to identify, $\boldsymbol{l}$.

2) Construct the data matrices:

$$\boldsymbol{Y} = [\boldsymbol{y_0}\ \boldsymbol{y_1} \cdots \boldsymbol{y_l} \cdots \boldsymbol{y_m}] \tag{5a}$$

$$\text{and } \boldsymbol{V} = \begin{bmatrix} \boldsymbol{u_0} & \boldsymbol{u_1} & \cdots & \boldsymbol{u_l} & \cdots & \boldsymbol{u_m} \\ \boldsymbol{0} & \boldsymbol{v_0} & \cdots & \boldsymbol{v_{l-1}} & \cdots & \boldsymbol{v_{m-1}} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \boldsymbol{0} & \boldsymbol{0} & \cdots & \boldsymbol{v_0} & \cdots & \boldsymbol{v_{m-l}} \end{bmatrix}, \tag{5b}$$

where $\boldsymbol{v_i} = [\boldsymbol{u_i^T}\ \boldsymbol{y_i^T}]^{\boldsymbol{T}}$ represents the concatenated input-output data at time i. The matrix $\boldsymbol{V}$ resembles $\mathbf{U}$, except that it has been augmented with the outputs y_i. In this way, we are working with a system that is augmented to include a Kalman filter. We are now identifying the observer Markov parameters of the augmented system, $\bar{\mathbf{Y}}^\delta$, using (3). It will be possible to identify these observer Markov parameters from the data and then extract the impulse response (Markov parameters) of the original system.

3) Identify the matrix $\bar{\boldsymbol{Y}}^{\boldsymbol{\delta}}$ of observer Markov parameters by solving $\boldsymbol{Y} = \bar{\boldsymbol{Y}}^{\boldsymbol{\delta}}\boldsymbol{V}$ using the right pseudo inverse of $\boldsymbol{V}$ (i.e. SVD).

4) Order the observer Markov parameters $\bar{\boldsymbol{Y}}^{\boldsymbol{\delta}}$ as:

$$\bar{\boldsymbol{Y}}_{\boldsymbol{0}}^{\boldsymbol{\delta}} = \boldsymbol{D},\ \bar{\boldsymbol{Y}}_{\boldsymbol{k}}^{\boldsymbol{\delta}} = [(\bar{\boldsymbol{Y}}^{\boldsymbol{\delta}})_{\boldsymbol{k}}^{(\boldsymbol{1})}\ (\bar{\boldsymbol{Y}}^{\boldsymbol{\delta}})_{\boldsymbol{k}}^{(\boldsymbol{2})}] \text{ for } \boldsymbol{k} \geq \boldsymbol{1}. \tag{6}$$

where $(\bar{Y}^\delta)_k^{(1)} \in \mathbb{R}^{q\times p}$, $(\bar{Y}^\delta)_k^{(2)} \in \mathbb{R}^{q\times q}$, and $\boldsymbol{y_0^\delta} = \bar{\boldsymbol{Y}}_{\boldsymbol{0}}^{\boldsymbol{\delta}} = \boldsymbol{D}$.

5) Reconstruct system Markov parameters:

$$y_k^\delta = (\bar{Y}^\delta)_k^{(1)} + \sum_{i=1}^{k} (\bar{Y}^\delta)_i^{(2)} y_{k-i}^\delta \text{ for } k \geq 1 \tag{7}$$

Once the impulse response for an arbitrary input-output configuration is obtained using OKID, ERA can be used as discussed in the next to obtain the state space matrices.

The OKID implementation operates on input/output data $\mathbf{u} \in \mathbb{C}^{m\times l}$ and $\mathbf{y} \in \mathbb{C}^{q\times l}$ using an observer horizon p. Its time complexity is

$$\begin{aligned} T(m,q,l,p) = \mathcal{O}\big((m+q)\,p\,l\big) + \ \mathcal{O}\big(q^2 m\, p^2\big) \\ + \ \mathcal{O}\big(\min\big((m+q)p,\ l\big)\,(m+q)p\,l\big). \end{aligned} \tag{8}$$

which, for the standard choice $p = \Theta(\min(m,q)\,l)$, simplifies to $T = O\big((m+q)\min(m,q)\,l^3\big)$. The input data matrix is denoted by $\mathbf{u} \in \mathbb{C}^{m\times l}$, where m represents the number of inputs (rows) and l denotes the number of time samples (columns). Similarly, the output data matrix is $\mathbf{y} \in \mathbb{C}^{q\times l}$, where q is the number of outputs (rows) and l again represents the number of time samples (columns). The parameter p denotes the number of observer Markov parameters identified; it is typically chosen to be four to five times the effective system order and can be expressed as $p = \Theta(\min(m \times l,\ q \times l))$.

b) State-Space Realization with ERA: With the estimated Markov parameters $\mathbf{y}_k^\delta$ from OKID, we use the Eigensystem Realization Algorithm (ERA) to find a minimal-order state-space model $(\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D})$ [18], [19]. ERA begins by arranging the Markov parameters into a Hankel matrix:

$$\mathbf{H} = \begin{bmatrix} \mathbf{y}_1^\delta & \mathbf{y}_2^\delta & \cdots & \mathbf{y}_m^\delta \\ \mathbf{y}_2^\delta & \mathbf{y}_3^\delta & \cdots & \mathbf{y}_{m+1}^\delta \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{y}_n^\delta & \mathbf{y}_{n+1}^\delta & \cdots & \mathbf{y}_{m+n-1}^\delta \end{bmatrix} \tag{9}$$

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The ERA procedure requires the singular value decomposition (SVD) of the Hankel matrix:

$$\mathbf{H} = \mathbf{P}\boldsymbol{\Sigma}\mathbf{Q}^* = [\tilde{\mathbf{P}}\ \mathbf{P_t}] \begin{bmatrix} \tilde{\boldsymbol{\Sigma}} & 0 \\ 0 & \boldsymbol{\Sigma}_\mathbf{d} \end{bmatrix} [\tilde{\mathbf{Q}}^*\ \mathbf{Q}_\mathbf{t}^*] \approx \tilde{\mathbf{P}}\tilde{\boldsymbol{\Sigma}}\tilde{\mathbf{Q}}^* \quad (10)$$

The small singular values in $\boldsymbol{\Sigma}_\mathbf{d}$ are discarded, retaining only the first r singular values in $\tilde{\boldsymbol{\Sigma}}$. The columns of $\tilde{\mathbf{U}}$ and $\tilde{\mathbf{V}}$ form the eigen-time delay coordinates.

ERA computes the reduced-order system matrices $(\tilde{\mathbf{A}}, \tilde{\mathbf{B}}, \tilde{\mathbf{C}}, \tilde{\mathbf{D}})$ that approximate the system's dynamics based on the matrices $\mathbf{H}$ and $\mathbf{H}'$, [18], [20]. $\mathbf{H}'$ is a time-shifted version of $\mathbf{H}$, defined as:

$$\mathbf{H}' = \begin{bmatrix} \mathbf{y}_2^\delta & \mathbf{y}_3^\delta & \cdots & \mathbf{y}_{m+1}^\delta \\ \mathbf{y}_3^\delta & \mathbf{y}_4^\delta & \cdots & \mathbf{y}_{m+2}^\delta \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{y}_{n+1}^\delta & \mathbf{y}_{n+2}^\delta & \cdots & \mathbf{y}_{m+n}^\delta \end{bmatrix} \quad (11)$$

In ERA, the parameters m and n specify the block dimensions of the Hankel matrix H, which directly determines the quality of the reduced-order state-space realization. The order of the model is determined by the number of significant singular values retained from the SVD.

In the present implementation, the block dimensions m and n used to assemble the Hankel matrix $\mathbf{H}$ are chosen automatically from the Hankel SVD (see Section II). Concretely, after computing the OKID-obtained Markov-parameter tensor H_{mkv} we form a block-Hankel matrix H_0, compute its singular values $\{\sigma_i\}$, and declare a singular value significant if its magnitude relative to the largest singular value exceeds a threshold (i.e. $\sigma_i/\sigma_1 > \tau$). The resulting count n_{sig} of significant modes is then used to set the Hankel block size via $m = n = \left\lceil \sqrt{n_{\mathrm{sig}}} \right\rceil$. This systematic selection produces a compact Hankel realization of size $m \times n$ (total system order of n_{sig}), that reflects the system's intrinsic dynamic order and is robust to measurement noise, avoiding manual tuning.

The identified matrices are:

$$\tilde{\mathbf{A}} = \tilde{\boldsymbol{\Sigma}}^{-1/2}\tilde{\mathbf{P}}^*\mathbf{H}'\tilde{\mathbf{Q}}\tilde{\boldsymbol{\Sigma}}^{-1/2} \quad (12a)$$

$$\tilde{\mathbf{B}} = \tilde{\boldsymbol{\Sigma}}^{1/2}\tilde{\mathbf{Q}}^* \begin{bmatrix} \mathbf{I}_{N_T} & 0 \end{bmatrix}^T \quad (12b)$$

$$\tilde{\mathbf{C}} = \begin{bmatrix} \mathbf{I}_{N_R} & 0 \end{bmatrix} \tilde{\mathbf{P}}\tilde{\boldsymbol{\Sigma}}^{1/2}, \text{ and } \tilde{\mathbf{D}} = \mathbf{y}_0^\delta. \quad (12c)$$

The ERA implementation constructs Hankel matrices of size $a \times b$ with $a = m\, n_{\mathrm{out}}$ and $b = n\, n_{\mathrm{in}}$, performs an SVD of H_1, and forms reduced matrices of rank r, which is the rank of the SVD matrix. Its time complexity is

$$T = O\big(mnn_{\mathrm{out}}n_{\mathrm{in}}\big) \ + \ O\big(\min(a,b)\, a\, b\big) \ + \ O\big(rab + r^2 b\big),$$

where the terms are, respectively, the Hankel assembly, the SVD/pseudo-inverse cost (dominant for rectangular SVD), and the matrix multiplications to form A, B, C. The variable n_{in} denotes the number of inputs, corresponding to the number of columns per block in the data, while n_{out} represents the number of outputs, corresponding to the number of rows per block. The parameters m and n define the Hankel block dimensions used to construct the matrices H and H', specifying the number of block rows and block columns, respectively. Finally, r denotes the rank of the singular value decomposition (SVD) matrix.

III. Results and Analysis

The system identification pipeline was implemented in MATLAB to model representative MIMO fading channels and to evaluate the fidelity of the identified state–space model (OKID–ERA) against simple, well-known per-sample equalizers. The received signal $\mathbf{y}_k$ is generated (memoryless, per-sample mapping) by applying the instantaneous channel matrix $\mathbf{W}_k$ to the transmitted vector $\mathbf{u}_k$, with additive white Gaussian noise (AWGN) η_k (for time k):

$$\mathbf{y}_k = \mathbf{W}_k\mathbf{u}_k + \eta_k. \quad (13)$$

To provide baseline comparisons for the OKID–ERA identified model we used two standard equalization techniques-Zero-Forcing (ZF) and Minimum Mean-Square Error (MMSE) [21] and [22]. ZF and MMSE are standard, easily interpretable single-shot baselines for MIMO systems. They operate on the instantaneous channel $\mathbf{W}_k$ and are optimal under simple linear/ Gaussian assumptions (ZF for interference nulling, MMSE for MSE minimization).

In addition to these classical baselines, we also implemented a deep learning (DL) benchmark inspired by recent works on data-driven channel estimation [23]–[25]. The DL model was trained to learn the mapping from noisy received symbols and instantaneous channel state information to least-squares (LS) based symbol estimates. The input features to the network consisted of the real and imaginary parts of the power-normalized received vector $\mathbf{y}_k$, the flattened real and imaginary parts of the channel matrix $\mathbf{W}_k$, and a normalized scalar encoding the operating SNR. The training targets were the corresponding symbol estimates, also normalized for stability, such that the network learned to approximate the LS solution across a wide SNR range. The architecture comprised a feature input layer followed by four fully connected hidden layers of sizes 640, 512, 256, and 128 neurons, each with ReLU activations. Dropout regularization (rates 0.30, 0.25, and 0.20) was inserted after the first three hidden layers to improve robustness against overfitting. The output layer predicted the real and imaginary parts of the transmitted symbol vector with a regression loss function. Training employed the Adam optimizer with an initial learning rate of 5×10^{-4}, a mini-batch size of 256, and a maximum of 60 epochs, with piecewise learning-rate decay, standardized inputs, and validation-based early stopping.

A. Noisy One-Ring Model with Doppler Shift

We first consider the standard one-ring scattering model [26]. The time-varying channel matrix $\mathbf{W}_k$ is modeled as

$$\mathbf{W}_k \ = \ \frac{1}{\sqrt{L}} \sum_{l=1}^{L} \epsilon_l\, \mathbf{a}_{\mathrm{MS}}(\theta_l)\, \mathbf{a}_{\mathrm{BS}}^*(\phi_l)\, e^{j2\pi f_d k \Delta t \cos\theta_l}, \quad (14)$$

where L is the number of scatterers, ϵ_l are complex gains, $\mathbf{a}_{\mathrm{MS}}$ and $\mathbf{a}_{\mathrm{BS}}$ are array response vectors, f_d is the maximum Doppler shift, and Δt is the time step. The Hermitian transposition operation is denoted by $*$. The simulation configuration used a carrier frequency $f_c = 5$ GHz, $N_T = 8$ transmitters, $N_R = 15$ receivers and a mobile speed of 30 m/s. The system

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response was generated for $n = 230$ time samples. A key advantage of the proposed framework is its robustness to noise. We evaluated the identification performance across a range of SNRs: -50, -20, 20, and 50 dB.

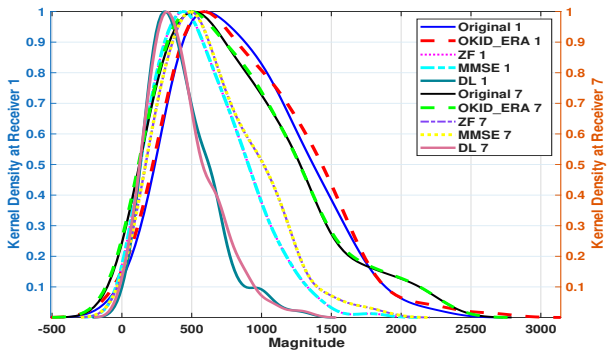

Fig. 1: Kernel densities of the outputs at receivers 1 and 7 for the noisy One-Ring (OR) channel at SNR= -50 dB.

TABLE I: Relative error (%) vs SNR (One-Ring model)

SNR	-50 dB	-20 dB	20 dB	50 dB
OKID–ERA	8.2066	5.6424	3.4333	3.3120
ZF	29.0596	28.8371	1.7018	0.0550
MMSE	29.0580	28.8315	1.7018	0.0550
DL	32.6360	33.1028	7.8885	7.1323

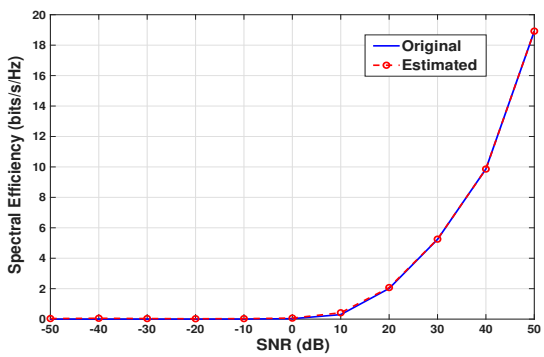

Fig. 2: Spectral efficiency of the noisy One-Ring channel.

Fig. 1 shows representative tracking capability at -50 dB SNR for the noisy OR channel model. Table I lists the relative error (%) (magnitude-based) for the OKID–ERA identified model, the instantaneous ZF / MMSE reconstructions, and the DL model. For low SNR (noisy) regime, e.g. -50 and -20 dB SNRs, OKID–ERA substantially outperforms the instantaneous equalizers. The relative error of the OKID–ERA identified output is roughly $5-8\%$, whereas ZF, MMSE and DL produce errors of $\approx 30\%$. This demonstrates that, when observations are heavily contaminated by noise, constructing a dynamical model from the measured data provides strong de-noising and filtering benefits: OKID-ERA fits a reduced-order state–space model that captures temporal correlations and averages out measurement noise.

Next, the spectral efficiency of the MIMO link is compared to quantify how well the identified model preserves information-theoretic properties of the channel. For an $N_T \times N_R$ narrow-band MIMO channel $\mathbf{W_k}$ with total transmit power P and noise variance σ^2, we use the standard expression [22], [27]:

$$C(\mathbf{W_k}) \;=\; \log_2 \det\left(\mathbf{I}_{N_R} + \frac{P}{N_T\sigma^2}\,\mathbf{W_k}\mathbf{W_k}^*\right) \text{ (bps/Hz)}. \tag{15}$$

For the state-space identified channel using OKID-ERA, the input/output pair $(\mathbf{u_k}, \mathbf{y_k})$ of are applied in the OKID/ERA technique to estimate the state matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$ and $\mathbf{D}$ which are used to estimate the equivalent MIMO channel as:

$$\mathbf{W}_{\text{est}} \;=\; \mathbf{D} \;+\; \mathbf{C}\,(\mathbf{I}-\mathbf{A})^{-1}\mathbf{B}, \tag{16}$$

It follows directly by evaluating the state–space transfer function $\mathbf{W}(z) = \mathbf{D} + \mathbf{C}(z\mathbf{I}-\mathbf{A})^{-1}\mathbf{B}$ at $z = 1$ (see e.g. [16], [28]). The formula is valid provided $(\mathbf{I}-\mathbf{A})$ is invertible (equivalently, $z = 1$ is not an eigenvalue of A); if this condition is not met, one should take the appropriate frequency-response limit. Here, $\mathbf{I}$ is the identity matrix. Fig. 2 shows the spectral efficiency curves obtained from the true instantaneous channel (time-averaged) and from the identified OKID–ERA model with close similarity.

B. Noisy Rician Fading with LoS and Stochastic Paths

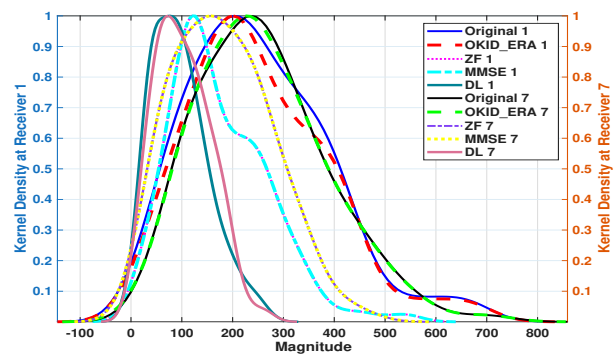

Fig. 3: Kernel densities of the outputs at receivers 1 and 7 for the noisy Rician channel at SNR= -50 dB.

Next, we evaluate a Rician fading environment which contains a deterministic Line-of-Sight (LoS) component together with randomly scattered multipath contributions [26]. The time-varying channel matrix is modeled as

$$\mathbf{W}_k = \sqrt{\frac{K}{K+1}}\,\mathbf{W}_{\text{LoS}} + \sqrt{\frac{1}{K+1}}\,\mathbf{W}_{\text{scat}}, \tag{17a}$$

$$\mathbf{W}_{\text{LoS}} = \mathbf{a}_{\text{MS}}(\phi_{\text{LoS}})\,\mathbf{a}^*_{\text{BS}}(\theta_{\text{LoS}}), \tag{17b}$$

$$\mathbf{W}_{\text{scat}} = \frac{1}{\sqrt{L}}\sum_{l=1}^{L}\epsilon_l \mathbf{a}_{\text{MS}}(\phi_l)\mathbf{a}^H_{\text{BS}}(\theta_l)e^{j2\pi f_d k\Delta t\cos\theta_l}, \tag{17c}$$

where K is the Rician K-factor and notation follows (14). For this experiment, we used $N_T = 8$, $N_R = 15$, a Rician factor $K = 10$, and generated $n = 230$ time samples of the memoryless, time-varying channel.

The distribution of the received signal is compared for the Rician channel in Fig. 3. The channel spectral efficiency is compared in Fig. 4. In both Figs. 3 and 4, the identified channel's response matches closely with the actual system

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response. The performance of the proposed OKID-ERA-based channel estimation is compared against ZF, MMSE, and DL in Table II, where the OKID/ERA outperforms the other three techniques in the low SNR regime.

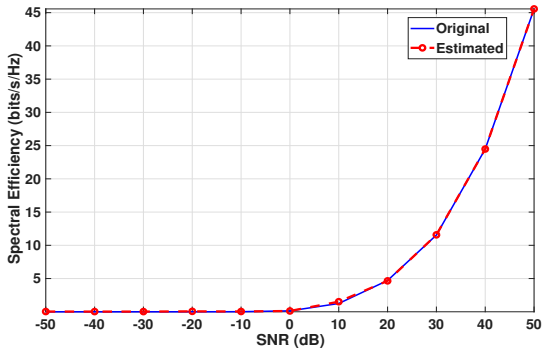

Fig. 4: Spectral efficiency of the noisy Rician channel.

TABLE II: Relative error (%) vs SNR (Rician model)

SNR	-50 dB	-20 dB	20 dB	50 dB
OKID–ERA	7.0211	6.8244	3.9323	3.9493
ZF	29.3029	29.0002	1.7255	0.0548
MMSE	29.3027	29.0003	1.7255	0.0548
DL	48.7005	46.5841	8.1195	7.8500

In short, these experiments support the claim that OKID–ERA is a robust identification method in noisy measurement scenarios. It *substantially* outperforms the instantaneous inversion baselines (ZF and MMSE) and DL at low SNR by leveraging temporal structure and model-based regularization.

IV. Conclusion

This paper introduced a data-driven framework using OKID and ERA for modeling MIMO communication systems. Unlike data-hungry AI/ML methods or complex analytic techniques, the proposed approach successfully constructs low-dimensional, predictive state-space models with a limited set of input-output data for accurately capturing the channel dynamics. The results demonstrate that the proposed control-theoretic method achieves high fidelity in modeling system response and predicting channel spectral efficiency for different fading environments. Furthermore, the framework shows exceptional tolerance to noise, maintaining its effectiveness for SNR values as low as -50 dB. This work highlights the potential of system identification techniques for developing efficient and robust models for the next generation of wireless systems.

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